RENAD ALANAZI

Electrical Engineer · Embedded Systems · PCB Design · Instrumentation

Riyadh, Saudi Arabia • +966 53 046 0936 • alanazirenad.f@gmail.com • linkedin.com/in/renad-alanazi • github.com/Renad-Alanazi • arcvolitx.com

Saudi National · Arabic (Native), English (Fluent)

PROFESSIONAL SUMMARY

Electrical engineer (B.S., Boston University, 2026) based in Riyadh with hands-on experience in PCB design, embedded firmware, and sensor instrumentation, with laboratory work in semiconductor fabrication and cleanroom processing. Able to take hardware from schematic capture and multilayer layout through firmware, board bring-up, and bench validation. Seeking an entry-level electrical, instrumentation, or controls engineering role with a multinational operator or EPC in energy, oil & gas, or industrial automation; open to field assignments across the Kingdom.

CORE TECHNICAL SKILLS

- **PCB & Hardware Design:** Multilayer PCB layout, schematic capture, analog and digital circuit design, design-for-manufacturing (DFM), Gerber and BOM generation, prototyping, hardware bring-up and debug
- **Embedded Systems & Firmware:** Embedded C/C++; nRF52840, ESP32, ARM Cortex-M; low-power firmware; BLE and Wi-Fi connectivity; sensor interfacing and hardware-software integration
- **Power & Instrumentation:** Power-subsystem and power-electronics design, wireless power transfer, battery and low-power optimization, signal acquisition and conditioning, IMU/accelerometer and analog/digital sensor interfacing
- **Test & Validation:** Oscilloscope, multimeter, function generator, logic analyzer; soldering and rework; functional and design verification; root-cause analysis
- **Semiconductor & Cleanroom:** IC CAD layout, photolithography, wet etching, thin-film deposition; cleanroom processing and contamination control
- **Software & Tools:** MATLAB, Python, LTspice, Cadence Virtuoso, Altium Designer, EasyEDA, Git/GitHub; React, Next.js, TypeScript, Tailwind CSS, Supabase

PROFESSIONAL EXPERIENCE

ArcVolitX | PCB Design Engineer — Riyadh, Saudi Arabia · Feb 2026 – Present

- Design multilayer PCBs in EasyEDA from schematic capture and component selection through DFM review, then generate Gerber/BOM packages.
- Build ESP32- and nRF52840-based embedded systems for wireless sensing, instrumentation, and control, integrating analog and digital sensors over BLE/Wi-Fi with embedded C/C++ firmware.
- Bring up boards and debug hardware/firmware faults to root cause using an oscilloscope, multimeter, logic analyzer, and function generator.
- Prepare schematics, test procedures, and design notes for review and fabrication handoff.

Boston University, College of Engineering — Boston, MA · Sep 2023 – May 2025

Professor's Assistant | EC578 Fabrication Technology of Integrated Circuits — Jan 2025 – May 2025

- Supported wafer-fabrication projects in the cleanroom across IC CAD layout, photolithography, etching, and thin-film deposition; guided students and assisted in lectures.

Teaching Assistant | EC410 Introduction to Electronics — May 2024 – Dec 2024

- Guided students through electronics analysis and lab work on diodes, transistors, and amplifiers, debugging their breadboarded designs.

Teaching Assistant | EK307 Electric Circuits — Sep 2023 – May 2024

- Coached students on circuit analysis and measurement, diagnosing faults with oscilloscopes, multimeters, and function generators to isolate root cause.

Lab Monitor | College of Engineering — May 2024 – May 2025

- Kept lab and test equipment in safe working order and served as first point of contact for hands-on troubleshooting.

ENGINEERING PROJECTS

Personal Alert Device (PAD) | Embedded Wearable Monitoring System

- Built a wearable emergency and health-monitoring device on the Seeed XIAO nRF52840 Sense (ARM Cortex-M4F, onboard 6-axis IMU) over BLE.
- Designed a custom 2-layer PCB from schematic capture through layout, generating Gerber and BOM files for fabrication.
- Integrated the MAX30102 optical sensor for heart-rate monitoring and used the onboard IMU for motion sensing and fall detection.
- Developed embedded C/C++ firmware for sensor acquisition, signal processing, fall detection, and BLE communication with a mobile app.
- Designed the power architecture around an 1100 mAh lithium-ion battery with an integrated wireless charging subsystem for daily portable use.
- Ran bring-up, bench testing, and validation across board revisions, troubleshooting hardware and firmware faults to root cause and improving reliability through iterative testing.

Wireless Charging System for Wearable Electronics

- Designed an inductive wireless power transfer system with a custom charging dock and transmitter/receiver coil architecture, delivering a measured 5.07 V and 0.487 A (2.47 W) at the receiver during bench validation.

Semiconductor Fabrication & Photolithography

- Fabricated Ti/Au microstructures through a full cleanroom workflow: CAD layout, photolithography, thin-film metal deposition, and wet etching, with wafer inspection and characterization.

Automotive Rain-Sensing Control System

- Built a sensor-driven system that auto-activates headlights and wipers via MOSFET- and relay-based power switching; bench-tested and validated.

Discrete-Logic Traffic Signal Controller

- Built a complete traffic-intersection signal controller using only discrete hardware: NE555 timing circuits, CMOS logic, analog comparators, and MOSFET drivers, with battery-powered operation and no microcontroller or firmware.

EDUCATION

B.S. in Electrical Engineering, Boston University | Boston, MA · 2026

Relevant coursework: Circuit Analysis · Electronics · Signals & Systems · Control Systems · Semiconductor Devices